

Emora: Implementation of a Local-First, Emotion-Aware AI Companion with Selective Memory and Multimodal Presence

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Abstract—Emotion-aware companions require more than fluent text generation: they need continuity, restrained affective adaptation, user control, and clear boundaries. This paper reports the completed implementation of Emora, a browser-based AI companion whose primary inference path runs locally on Apple Silicon. The system combines a quantized Qwen3 language model, an explainable text-emotion estimator, optional one-frame Qwen2-VL visual check-ins, selective persistent memory, deterministic personality and behavior planning, Qwen3-TTS with a Kokoro fallback, and a WebGL VRM avatar. A FastAPI application coordinates authenticated conversation state and MongoDB persistence. The implementation deliberately does not fine-tune or train a new model. Instead, it constrains pretrained inference with bounded context, trusted-data prompt separation, allowlisted avatar actions, clamped behavior parameters, explicit privacy controls, and user-owned deletion and export paths. Repository verification produced 169 passing automated tests and one deliberately skipped live-MongoDB integration test. Existing local benchmark records show functional model and audio pipelines, but they do not establish emotional recognition accuracy, therapeutic benefit, or user acceptance. We therefore separate software evidence from a proposed human evaluation protocol. The principal contribution is a reproducible systems integration that converts a conceptual companion framework into an inspectable application while making its unverified affective claims and safety limitations explicit.

Index Terms—affective computing, conversational agents, AI companions, local inference, selective memory, multimodal interaction, human-AI interaction, responsible AI

I. INTRODUCTION

Conversational systems have progressed from scripted pattern matching to transformer-based generation capable of coherent, open-domain dialogue [1], [2]. Fluency alone, however, is insufficient for a companion intended for repeated interaction. Users reasonably expect the system to remember relevant facts, maintain a recognizable identity, respond appropriately to emotional language, and communicate its limitations. Gaps between such expectations and actual agent capability can rapidly undermine trust [3].

Affective computing studies how computational systems can recognize, represent, and respond to affective signals [4]. Empathetic dialogue research similarly shows that emotional situations require more than generic conversational data and that human judgment is central to evaluating response empathy [5].

These goals become especially sensitive in companion applications. Warm language may be helpful, yet anthropomorphic presentation, persistent memory, and emotional adaptation can also create privacy, over-reliance, and misinterpretation risks [6], [7].

Our earlier project manuscript proposed a “Stability-First” methodological framework involving probabilistic emotion inference, temporal smoothing, candidate-response scoring, curriculum learning, and constrained optimization. The completed software follows a more conservative and auditable design. It uses pretrained models at inference time and surrounds them with deterministic application logic. Text affect is estimated through an explicit lexicon; visual input is optional and momentary; durable memory is extracted only from recognizable user statements; the model receives bounded recent history and trusted context; and a post-generation Companion Brain produces clamped, allowlisted behavior for speech and avatar animation.

This paper makes three contributions. First, it reconstructs the implemented end-to-end architecture from the source repository rather than presenting an aspirational design. Second, it documents how emotion, memory, personality, voice, and avatar behavior are connected without mischaracterizing inference as training. Third, it reports available software and local-runtime evidence while separating it from emotional-quality and human-subject claims that have not yet been evaluated.

II. RELATED WORK

A. Affective and Multimodal Computing

Picard established affective computing as a technical and human-centered research area [4]. Discrete emotion taxonomies, including Ekman’s account of basic emotions, influenced many classification systems [8]; nevertheless, lived affect is contextual and cannot be reduced safely to a label. Reviews of affective computing describe text, speech, facial, and multimodal signals along with the limitations introduced by ambiguous expression, culture, environment, and dataset bias [9]. A meta-analysis of multimodal affect detection found that fusion can improve performance but also emphasized variation in experimental design and generalizability [10].

Emora therefore labels its outputs as momentary, non-clinical estimates and retains an explicit “unclear” visual state.

B. Empathetic and Social Dialogue

Long-running social chatbots highlight the engineering challenge of balancing engagement, personalization, and consistent persona [2]. The EmpatheticDialogues benchmark demonstrated the value of emotion-grounded data and human judgments for empathetic response generation [5]. Emora does not train on that dataset and does not claim comparable benchmark performance. It instead uses a general pretrained LLM with application-level prompts and context. This distinction matters: an emotionally worded response is not evidence that the system inferred the user’s state accurately or behaved empathetically.

C. Human–AI Interaction and Ethics

Human–AI design guidance recommends making system capability visible, supporting correction, and helping users understand why a system acted as it did [11]. Responsible-AI literature further motivates autonomy, transparency, privacy, and non-maleficence [7]. Emora operationalizes a subset of these principles through opt-in camera access, editable memories, data export and deletion, non-diagnostic copy, and explicit AI-role language. These mechanisms reduce specific risks but do not constitute a complete safety guarantee.

III. PROBLEM STATEMENT AND DESIGN OBJECTIVES

Let a conversational turn contain user text x_t , optional image v_t , bounded recent history H_t , relevant persistent memories M_t , explicit preferences P_t , and character configuration C_t . The implemented system produces a textual response y_t and a presentation plan B_t for voice and avatar behavior:

$$(y_t, B_t) = F(x_t, v_t, H_t, M_t, P_t, C_t). \quad (1)$$

Unlike the earlier conceptual formulation, F is not trained end-to-end by this project and has no explicit emotional-transition loss. It is a composition of pretrained inference and deterministic application functions.

The implementation was guided by six objectives: (1) maintain conversational continuity without storing every utterance as durable memory; (2) use emotional cues conservatively and visibly as estimates; (3) keep character and presentation outputs bounded; (4) support typed, spoken, and optional visual interaction; (5) keep the primary AI path local and user-controlled; and (6) distinguish supportive conversation from therapy, diagnosis, crisis care, or a human relationship.

IV. SYSTEM ARCHITECTURE

Figure 1 shows the deployed logical structure. The browser presents server-rendered Jinja pages enhanced with vanilla JavaScript. It supports typed input, browser speech recognition when available, Web Audio playback, an optional camera capture, and a Three.js/VRM avatar. HTTPS/JSON requests enter FastAPI, which authenticates the user, applies rate and entitlement checks, owns conversation state, and assembles trusted context. MongoDB stores accounts, conversations, memories,

TABLE I
CONCEPTUAL CLAIMS COMPARED WITH THE IMPLEMENTATION

Concept	Status	Repository-grounded interpretation
Text emotion recognition Visual affect input	Implemented differently Partial	Deterministic 13-label lexicon; non-clinical estimate Opt-in single frame; six expression labels plus uncertainty
Temporal emotion smoothing	Not implemented	Stored per-turn signals support later summaries; no recurrent smoother
Long-term memory	Implemented	Selective facts in MongoDB with relevance ranking and user controls
Character consistency	Implemented differently	Stable prompts, four numeric profiles, clamping, and gesture allowlist
Stability-constrained decoding	Not implemented	One generated reply is post-processed; no candidate optimization
Curriculum/RL training	Not implemented	Pretrained quantized models are used without project fine-tuning
Ethical boundaries	Partial	Prompt rules, UI disclosure, consent, retention, export, and deletion; no independent chat moderation model
Formal efficacy study	Not performed	Software tests and local technical benchmarks only

preferences, user-generated content, and operational metadata. AI inference is selected through a provider interface: the default is local MLX on Apple Silicon [12], while explicitly configured local HTTP, Modal, or cloud-compatible endpoints may be used as deployment alternatives.

A. Turn Processing

A message is validated with a maximum application payload of 12,000 characters and assigned a client turn identifier. The API claims this identifier before generation to prevent duplicate replies. The user message, its text-emotion analysis, and a privacy-dependent behavior report are associated with the conversation. If a camera frame is present, both account preference and per-request opt-in are required; failure of visual inference does not block text chat.

The service retrieves recent conversation history and a small set of relevant memories. It then builds a system message containing the companion role, persona, strict output contract, trusted account data, preferences, interaction mode, optional visual observation, and—when deliberately enabled—bounded web evidence. Stable account facts such as the user’s profile name and plan can be answered by application-owned logic rather than regenerated by the model. On completion, the API attaches the assistant response to the originating turn with one MongoDB aggregation update, persists presentation metadata, and only then extracts new durable-memory candidates.

B. Concurrency and Degraded Operation

MLX model instances are loaded lazily and cached for the process lifetime. A generation lock serializes access to shared model state, while bounded queues enforce per-user concurrency and prioritize entitled requests. Provider health is cached briefly and the manager attempts enabled providers in

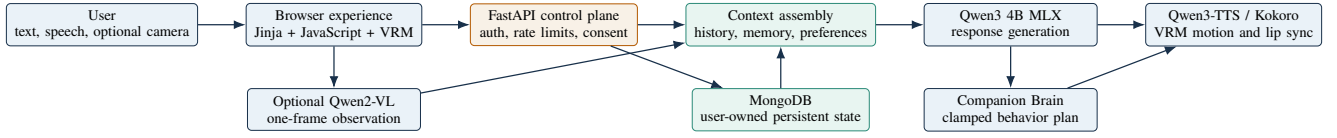


Fig. 1. Implemented architecture. Camera pixels are processed in memory for one request and are not stored. The provider layer is MLX-first; remote alternatives require explicit configuration.

priority order. A visual failure degrades to text; a voice failure may fall back from Qwen3-TTS to Kokoro; and an unavailable generation provider returns an explicit service error rather than a fabricated response.

V. AI/ML AND MULTIMODAL IMPLEMENTATION

Qwen3 provides the main generative capability and supports thinking and non-thinking modes [13]. Emora appends Qwen’s native directive and strips any leading private-reasoning block before output. A strict prompt requests only user-visible words, and a cleaning stage removes known non-companion claims and leading sad emoticons. This is inference-time control, not a guarantee that every unsafe or incorrect model output will be removed.

The vision model is prompted to return only visibility, expression, engagement, confidence, a short summary, and a support cue. Allowed expressions are *pleasant*, *neutral*, *tense*, *sad*, *surprised*, and *unclear*; engagement is *engaged*, *away*, or *uncertain*. Values outside these sets are replaced by uncertainty and confidence is clamped to $[0, 1]$. The prompt forbids identification and inference of age, gender, race, health, diagnosis, personality, or internal emotion. Thus the output is treated as a visible-cue report rather than ground truth about feeling.

VI. EMOTION-AWARE INTERACTION

A. Explainable Text Estimation

For normalized input x , the text analyzer counts phrase matches from a fixed set L_e for each emotion category e :

$$s_e(x) = \sum_{w \in L_e} \mathbb{1}[w \text{ occurs in } x], \quad \hat{e} = \arg \max_e s_e(x). \quad (2)$$

The implemented categories are happy, sad, excited, angry, lonely, nervous, calm, frustrated, embarrassed, confused, hopeful, motivated, and curious. If no phrase matches, a question becomes curious and other input becomes calm. Intensity follows the repository rule

$$I(x) = \min\{1, 0.35 + 0.22s_{\hat{e}}(x) + 0.08 \min(n, 3)\}. \quad (3)$$

The result also maps to fixed reflective metrics used by dashboards. These numbers are presentation signals, not probabilities, clinical scores, or validated measurements.

B. Influence on the Response and Avatar

The current text estimate and optional visual cue are inserted into the trusted context as data, with instructions not to diagnose or treat the cue as certain. The model is asked to reflect one concrete detail before offering a thought or question.

After generation, the same text estimate selects a coarse avatar posture such as comforting, focused, happy, curious, or thoughtful. The Companion Brain derives valence, arousal, curiosity, confidence, empathy, engagement, and dominance values from the user message, reply, and character profile. Every value is clamped; gesture requests are restricted to an allowlist; and sentence boundaries determine a declarative behavior timeline.

No temporal emotion filter, probabilistic fusion model, facial-expression classifier accuracy study, or emotional-transition constraint is implemented. Per-turn analyses persist with conversation messages and may be aggregated for user-facing insights, but they do not form a learned recurrent affect state. When evidence is absent or visual analysis fails, text conversation continues with conservative defaults.

VII. MEMORY AND PERSONALITY MANAGEMENT

A. Short-Term and Persistent Context

Short-term continuity uses a bounded suffix of user and assistant messages. The service default is 16 messages; access policy and deep interaction mode can alter the supplied bound. This prevents an unbounded transcript from entering every inference request.

Persistent memory is selective. Regular-expression rules recognize explicit identity statements, preferences, work or education facts, pet relationships, goals, routines, and time-sensitive reminders. Vague conversation yields no durable candidate. Memories are upserted per user, category, and key. A changed identity, life, or relationship value is held as a pending conflict rather than silently overwriting the existing record. Temporary reminders expire after 28 days. Users can list, edit, delete, teach, and export memory-related data through authenticated routes.

Retrieval first loads at most 80 non-expired records ordered by importance and recency. Given message-word set W_x and memory-word set W_m , implemented relevance is

$$r(m, x) = \text{importance}(m) + \min\{0.7, 0.35|W_x \cap W_m|\}. \quad (4)$$

The highest-ranked records—normally at most eight—are supplied as data and marked as used. This is lexical retrieval, not embedding search, and may miss paraphrases or rank unrelated high-importance facts.

B. Character Consistency

Four presented companions—Yuna, Vivi/Rose, Sakurada/Robert, and Haru—have explicit natural-language personas in the room client. The server maps them to four

TABLE II
VERIFIED AI AND PRESENTATION COMPONENTS

Component	Implemented model/technology	Input → output	Integration and limits
Dialogue generation	Qwen3-4B, 4-bit MLX checkpoint [13]	Prompted messages → reply text	Pretrained; no project fine-tuning. Default temperature 0.7 and maximum 1024 generated tokens. Thinking is normally off and enabled automatically only for complex turns.
Visual check-in	Qwen2-VL-2B-Instruct, 4-bit MLX checkpoint [14]	One image → constrained JSON observation	Explicit opt-in; at most 2 MB, resized to 640 px maximum edge; pixels held in memory only.
Text affect	Deterministic repository lexicon	User text → label, intensity, signals	Thirteen categories; interpretable but not statistically calibrated and not trained.
Speech synthesis	Qwen3-TTS 12 Hz 1.7B CustomVoice, 6-bit MLX [15]	Reply plus style instruction → 24-kHz audio	Primary local engine with character voice mapping; sentence streaming and interruption support.
Voice fallback	Kokoro	Reply text → waveform	Local fallback when the primary runtime cannot load.
Speech input	Browser Web Speech API	Microphone audio → text	Availability and recognition behavior depend on the browser; not a repository-trained ASR model.
Embodied output	Three.js, pixiv three-vmr, Web Audio	Behavior plan and audio → motion, gaze, gestures, lip sync	Procedural animation; only allowlisted semantic gesture names reach the browser.

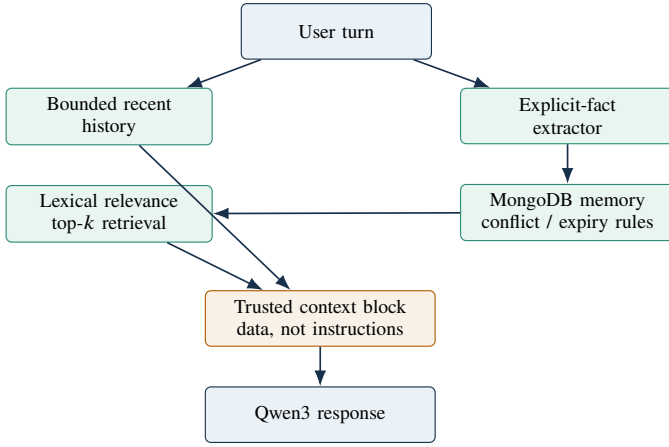


Fig. 2. Implemented continuity flow. Durable memory is selective and user-owned; recent history and retrieved facts remain separate inputs.

numeric profiles containing warmth, formality, humor, enthusiasm, energy, assertiveness, and curiosity. These profiles affect deterministic speech and behavior defaults, not the weights of the LLM. Conversation records persist character identity, persona prompt, and interaction mode. The shared system prompt additionally constrains response length, discourages repetitive greetings, limits follow-up questions, and prevents claims of human feelings or memory. Character consistency is therefore an application-level best effort, not a measured model property.

VIII. SAFETY, ETHICS, AND PRIVACY

Table III lists safeguards verified in code. The companion system identifies itself as AI, not a therapist or emergency service. For direct self-harm intent, the generation prompt instructs it to encourage local emergency services or a crisis line. It rejects possessive language, guilt, human-feeling claims, and uninvited “friend” language. Memory context is marked as untrusted data, and web evidence is marked as external evidence rather than instructions. Stable account operations such as plan changes and deletion cannot be executed by conversational text alone.

TABLE III
IMPLEMENTED SAFEGUARDS AND REMAINING BOUNDARIES

Area	Verified mechanism
Role clarity	AI/non-therapy disclosure in prompt and public trust surfaces
Dependence risk	No possessive, guilt, human-feeling, or uninvited-friend language in prompt
Crisis direction	Prompt requests emergency/crisis-line direction for imminent harm; no specialized classifier
Visual privacy	Separate preference and turn opt-in; one reduced frame; pixels not persisted
Data agency	Authenticated export, conversation/history deletion, memory editing/removal, account deletion
Authentication	SHA-256 pre-hash plus bcrypt passwords; expiring signed JWTs; revocable hashed sessions
Abuse resistance	Endpoint rate limits, entitlement checks, upload validation, security headers, production secret validation
Action safety	Chat cannot assert account mutation; avatar actions use bounded values and allowlisted gestures

Camera access is disabled by default. The browser captures a frame only with active permission when a message is sent. The server validates a data-image payload, limits decoded data to 2 MB, reduces the longest edge to 640 pixels, and does not write pixels to files, MongoDB, or logs. With data-minimization enabled, even the coarse visual report is excluded from the persisted behavior report. Emotional memory and adaptive context are separate user preferences; adaptive context is off by default.

Passwords use a SHA-256 pre-hash followed by bcrypt, and bearer sessions use signed JWTs with token versions and hashed server-side session tracking. Production startup rejects weak signing secrets, privileged routes require server-side authorization, and endpoints apply bounded rate limits. Security and privacy are nevertheless deployment properties as well as code properties: TLS termination, database access control, backups, email security, host hardening, and third-party configuration must be operated correctly.

The largest safety limitation is that companion chat has prompt-based behavioral constraints but no independent, validated crisis or moderation classifier. A generative model may ignore or incompletely follow instructions. The community feed has separate deterministic moderation and reporting con-

TABLE IV
EVIDENCE AVAILABLE AT MANUSCRIPT BUILD TIME

Question	Evidence	Defensible conclusion
Do application contracts execute?	169 passed, 1 skipped	Strong automated software evidence
Does local chat run?	Retained 4B benchmark	Functional on the measured Mac; not a broad latency study
Does TTS produce valid audio?	WAV health benchmark	File format and pipeline function; listening quality unscored
Is emotion recognition accurate?	No labeled test corpus	Unknown
Are replies empathetic and safe?	Prompt/unit checks only	Not established by human evaluation
Does long-term use improve well-being?	No longitudinal study	No claim can be made

trols, but those controls should not be generalized to private companion replies. Emora must therefore not be presented as a medical device, therapeutic intervention, or emergency resource.

IX. VERIFICATION AND EVALUATION

A. Completed Software Verification

On 1 September 2026, the repository compiled successfully with Python’s `compileall`. Pytest collected 170 tests: 169 passed in 1.76 s and one disposable-database integration test was skipped because the explicit `MONGO_INTEGRATION_TEST` flag was not enabled. The suite covers authentication and entitlements, public and protected routes, emotion and memory rules, vision-output validation, Companion Brain behavior, provider and queue behavior, TTS text and streaming contracts, community controls, web-search grounding, collaborative rooms, and workspace state. This result demonstrates internal software conformance under test doubles and local fixtures. It does not validate live MongoDB persistence, browser permissions, microphones, cameras, WebGL rendering, or human perception.

B. Recorded Local Technical Benchmarks

Repository benchmark records were produced on the development Apple Silicon machine. The selected Qwen3-4B MLX checkpoint showed warm scenario latency from 0.32 to 1.01 s and peak process resident memory of 2.52 GB; its recorded first scenario was 454.02 s because that run included the one-time model download. These figures are machine- and prompt-specific and are not general throughput claims. The retained TTS record reports a valid 24-kHz, 16-bit WAV of 12.56 s, a 9.3404 s first synthesis including model load, a 0.6 ms cache hit, and approximately 3.23 GB peak process memory. The 0.6 ms value measures retrieval of already cached audio, not fresh synthesis.

C. Proposed Human Evaluation

A future evaluation should be preregistered and reviewed for participant safety. It should compare Emora with an otherwise identical no-memory/no-affect condition using scripted and free conversation. A diverse participant sample should rate

response appropriateness, perceived empathy, persona consistency, transparency, and comfort. Evaluators should annotate false emotional assumptions, unwanted exclusivity, missed crisis escalation, privacy misunderstandings, and contradictions across sessions. Memory evaluation should measure factual precision, relevant recall, inappropriate recall, conflict handling, and user correction success. Camera evaluation requires consented, balanced data and separate reporting across lighting, occlusion, skin tone, expression ambiguity, and culture; the “unclear” rate should be treated as meaningful rather than as failure to force a label.

System evaluation should report cold and warm latency distributions, time to first audio, memory consumption with chat, vision, TTS, and VRM active together, interruption recovery, provider failures, and database concurrency. Human listening tests should score TTS naturalness, pronunciation, clipping, and whether style is recognizable without becoming theatrical. No deployment or well-being claim should be made until these studies are completed.

X. DISCUSSION

Emora is more than a basic stateless chatbot because it integrates authenticated continuity, selective user-owned memory, explicit emotion signals, optional visual observation, local language and speech inference, and embodied presentation. Its technically interesting feature is not a new foundation model but the boundary between probabilistic generators and deterministic product authority. Account facts, permissions, storage policy, action authorization, memory conflict handling, and avatar action names remain application-owned. The model contributes language generation and an optional constrained visual report; it does not control subscription state, deletion, or arbitrary browser functions.

This hybrid architecture makes several behaviors inspectable. The emotion estimator can explain which phrases matched. The memory selector exposes a simple relevance score. The Companion Brain clamps every continuous output and maps gestures to a fixed library. Camera pixels have a short, auditable lifetime. These properties trade statistical sophistication for controllability and are appropriate for an undergraduate systems project whose claims must be defensible from code.

The approach also reveals tensions. Lexicon matching is easy to audit but fragile under negation, sarcasm, spelling variation, mixed emotion, and multilingual text. Prompt constraints are flexible but weaker than independently enforced policies. Persistent memory improves continuity but increases the consequences of extraction errors. A warm avatar and voice can strengthen engagement while also increasing anthropomorphic interpretation. Responsible design therefore requires ongoing user control, conservative language, and evaluation of both helpfulness and over-reliance.

XI. LIMITATIONS

The implementation has not undergone a controlled user study, clinical evaluation, red-team campaign, or longitudinal

deployment study. There are no validated accuracy, precision, recall, F1, empathy, safety, bias, or well-being metrics. Automated tests mostly isolate components and cannot prove end-to-end behavior under real hardware, browser, network, database, or permission conditions.

Text emotion estimation is English-oriented and keyword-based. It has no negation model, calibration, cultural adaptation, or temporal smoothing. The visual model is general-purpose rather than a validated facial-affect classifier; its confidence is model-produced and merely clamped, not calibrated. The application does not infer vocal affect from speech. Browser speech recognition may involve browser-specific processing outside the local MLX path.

Memory retrieval uses lexical overlap and hand-written extraction patterns. It can miss paraphrases, save an imperfect span, or surface a high-importance but weakly related fact. Conversation and memory data are stored in MongoDB without application-level field encryption; protection depends on deployment, access control, and transport configuration. Local operation improves data locality for core inference, but optional web search, remote providers, CDN-hosted browser libraries, OAuth, and email are external when enabled.

Finally, character consistency is not formally measured, and prompt-based safety may fail. The system cannot substitute for professional care or emergency services. Its insights are reflective summaries of user-provided data, not measurements of mental state.

XII. FUTURE WORK

The immediate priority is evaluation rather than adding stronger affective claims. We plan to build a versioned conversation test set covering emotional ambiguity, memory conflicts, identity questions, dependency-seeking language, crisis cases, prompt injection, and culturally varied phrasing. Human raters should assess both supportive quality and appropriate restraint. A separate live integration environment should exercise MongoDB transactions, browser permissions, streaming audio, VRM frame rate, interruption, and complete account deletion.

Technical improvements may include multilingual and negation-aware affect estimation, calibrated uncertainty, privacy-preserving semantic memory retrieval, per-memory provenance in the UI, and an independent safety classifier with deterministic escalation templates. Any temporal affect model should be opt-in, bounded, explainable, and evaluated for the risk of carrying an incorrect interpretation across turns. On-device dependency pinning and local hosting of browser assets would further strengthen the local-first claim. Only after such evidence should adaptive personality or richer multimodal fusion be considered.

XIII. CONCLUSION

This paper documented Emora as implemented: a local-first AI companion that coordinates pretrained Qwen-family models with deterministic emotion estimation, selective persistent memory, stable character profiles, bounded behavior planning, local speech generation, and a VRM interface. The completed

system replaces the earlier conceptual training framework with an inspectable engineering pipeline. Its repository provides substantial automated software evidence and limited machine-specific runtime measurements, but no basis for claiming emotion-recognition accuracy, therapeutic benefit, or long-term emotional safety. The most defensible contribution is therefore a transparent integration architecture and an honest boundary around what remains to be evaluated. IEEEtran formatting is used only for manuscript presentation and does not imply review, acceptance, publication, or indexing by IEEE.

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